

## Report Pratica

### GameShell - Apprendimento della shell Linux tramite missioni

#### Obiettivo della pratica

L'obiettivo di questa esercitazione era prendere confidenza con l'interfaccia a riga di comando (shell) di un sistema Unix-like tramite GameShell, un gioco didattico che propone una serie di missioni per imparare a creare directory e file, cercare file secondo determinati criteri, eseguire e interrompere programmi, e altre operazioni standard da terminale.

#### Installazione delle dipendenze

Per prima cosa ho verificato/installato le dipendenze necessarie a GameShell tramite il gestore di pacchetti apt (gettext, man-db, procps, psmisc, nano, tree, bsdmainutils, x11-apps, wget), risultate già tutte presenti nel sistema:

```
(kali@kali)-[~]
$ sudo apt install gettext man-db procps psmisc nano tree bsdmainutils x11-apps wget
gettext is already the newest version (1.0-3).
man-db is already the newest version (2.13.1-1).
procps is already the newest version (2:4.0.6-3).
psmisc is already the newest version (23.7-2).
nano is already the newest version (9.2-1).
tree is already the newest version (2.3.2-1).
bsdmainutils is already the newest version (12.1.8).
x11-apps is already the newest version (7.7+11+b1).
wget is already the newest version (1.25.0-3).
The following packages were automatically installed and are no longer required:
  bluez-hcidump      libhdf5-310      libperl5.40      libx265-215      python3.13-dev
  firmware-marvell-presteria  libhdf5-hl-310  libpoppler156    network-manager-gnome  python3.13-minimal
  libabsl20260107  libbtrfs-3g89t64  libpython3.13-dev  perl-modules-5.40      python3.13-venv
  libbluray3      libopenjph0.27  libvidstab1.1     python3.13
Use 'sudo apt autoremove' to remove them.

Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 2
```

#### Download di GameShell

Al primo tentativo di scaricare lo script di avvio con wget ho ottenuto un errore di risoluzione del nome host, probabilmente dovuto a un problema temporaneo di connessione:



```
kali@kali: ~  
Session Actions Edit View Help  
Welcome to GameShell!  
Using and administering a computer running a Unix-like operating system (e.g., Linux, BSD or macOS) requires using a command line interface called "shell". In this game you will learn how to use the shell by journeying through various missions that will teach you to perform standard tasks such as:  
- creating directories,  
- creating, viewing and editing files,  
- searching for files satisfying certain criteria,  
- running and interrupting programs, additional flags are required if you want to run programs from inside GameShell. Refer to the section of the user manual.  
During your adventure you will need to visit locations, interact with people, and find various objects. However, you have to keep in mind that these are standard directories, programs, and files (often with random contents). So, for example:  
- "build a hut" means creating a directory called "Hut", and  
- "put a coin in your chest" means moving a file called "coin" to a directory called "Chest".  
This repository is configured to work with the Visual Studio Code Dev Containers extension. Press Enter to continue.
```

Il gioco spiega poi i comandi principali di gsh (il tool del gioco): gsh goal per scoprire l'obiettivo della missione corrente, gsh check per verificarne il completamento, e gsh help per l'elenco dei comandi disponibili. La posizione di partenza è la cartella World, che contiene le location Castle, Forest, Garden, Mountain e Stall:

```
---+---  
| README | GPL 3.0 license |  
---+---  
| Run the command |  
| $ gsh goal |  
| to discover your first mission. |  
| You can check the mission has been completed with |  
| $ gsh check |  
| The command |  
| $ gsh help |  
| displays the list of available (gsh) commands. |  
---+---  
|  
| Github Codespaces (or VSCode) |  
[mission 1] $ pwd  
/home/kali/gameshell/World  
[use 'gsh help' to get a list of available commands]  
[mission 1] $ ls  
Castle Forest Garden Mountain Stall
```

## Missioni completate

Di seguito riporto, nell'ordine, le missioni che ho documentato tramite screenshot.

### Raggiungere la cima della torre

Interrogando l'obiettivo con `gsh goal`, la missione richiedeva di raggiungere la cima della torre principale del castello, utilizzando i comandi `cd` (per spostarsi), `pwd` (per visualizzare la posizione corrente) e `ls` (per elencare i contenuti visibili):

```
Session Actions Edit View Help
| You can check the mission has been completed with |
| $ gsh check |
| |
| The command |
| $ gsh help |
| displays the list of available (gsh) commands. |
|-----|
| mission 1] $ gsh goal |
| |
| Mission goal |
|-----|
| Go to the top of the main tower of the castle. |
| (github Codespaces (or VSCode)) |
| Useful commands |
|-----|
| cd LOCATION |
| Move to the given location. |
| Remark: "cd" is an abbreviation for "change directory". |
| |
| pwd |
| Show the path to your current location. |
| Remark: "pwd" is an abbreviation for "print working directory". |
| |
| ls |
| Show a list of locations that are currently accessible. |
| Remark: "ls" is an abbreviation of "list". |
| |
| gsh check |
| Check if the mission objective has been achieved. |
| |
| gsh reset |
| Restart the mission from the beginning. |
| |
| Remarks |
|-----|
| UPPERCASE words appearing in commands are meta-variables: you need to replace them by |
| appropriate (string) values. |
| |
| Most filesystems treat uppercase and lowercase characters differently. Make sure you |
| use the correct path. |
|-----|
| (*)
```

Ho quindi navigato attraverso la struttura del castello (Castle → Main\_tower → First\_floor → Second\_floor → Top\_of\_the\_tower) e, una volta raggiunta la cima, ho verificato il completamento della missione con `gsh check`, ottenendo la conferma di successo:

```

[use 'gsh help' to get a list of available commands]
[mission 1] $ ls
Castle Forest Garden Mountain Stall

[use 'gsh help' to get a list of available commands]
[mission 1] $ cd Castle

[use 'gsh help' to get a list of available commands]
[mission 1] $ ls
Cellar Great_hall Main_building Main_tower Observatory

[use 'gsh help' to get a list of available commands]
[mission 1] $ cd Main_tower

[use 'gsh help' to get a list of available commands]
[mission 1] $ ls
First_floor

[use 'gsh help' to get a list of available commands]
[mission 1] $ cd First_floor

[use 'gsh help' to get a list of available commands]
[mission 1] $ ls
Second_floor

[use 'gsh help' to get a list of available commands]
[mission 1] $ cd Second_floor

[use 'gsh help' to get a list of available commands]
[mission 1] $ ls
Top_of_the_tower

[use 'gsh help' to get a list of available commands]
[mission 1] $ cd Top_of_the_tower

[use 'gsh help' to get a list of available commands]
[mission 1] $ gsh check

Congratulations, mission 1 has been successfully completed!

[ progress was saved in /home/kali/gameshell-save.sh ]

  |-----|
  | Use the command $ gsh help to get the list of "gsh" commands. |
  |-----|

```

## Esplorazione della cantina

Nella seconda missione mi sono spostato dalla cartella Castle alla sottocartella Cellar (la cantina), verificando la posizione con pwd ad ogni passaggio:

```

[use 'gsh help' to get a list of available commands]
[mission 2] $ cd Castle

[use 'gsh help' to get a list of available commands]
[mission 2] $ pwd
/home/kali/gameshell/World/Castle

[use 'gsh help' to get a list of available commands]
[mission 2] $ cd Cellar

```

Una volta raggiunta la location richiesta, il comando gsh check ha confermato il completamento della missione, con il progresso salvato automaticamente in un file di save:

```
[use 'gsh help' to get a list of available commands]
[mission 2] $ pwd
/home/kali/gameshell/World/Castle/Cellar

[use 'gsh help' to get a list of available commands]
[mission 2] $ gsh check
Congratulations, mission 2 has been successfully completed!

[ progress was saved in /home/kali/gameshell-save.sh ]
```

## Missione 6 - Spostare gli oggetti nel baule

In questa missione ho dovuto spostare tre monete (coin\_1, coin\_2, coin\_3) dal Giardino (Garden) al baule (Chest) situato nella Capanna della Foresta (Forest/Hut), utilizzando il comando mv per spostare più file contemporaneamente verso la destinazione:

```
~/Garden
[mission 6] $ mv ~/Garden/coin_1 ~/Garden/coin_2 ~/Garden/coin_3 ~/Forest/Hut/Chest/

~/Garden
[mission 6] $ gsh check
Congratulations, mission 6 has been successfully completed!

[ progress was saved in /home/kali/gameshell-save.sh ]

+-----+
| Use the command |
| $ gsh help      |
| to get the list of "gsh" commands. |
+-----+
```

## Missione 8 - Ripulire la cantina dai ragni

Nella cantina del castello ho trovato una grande quantità di file con nomi che rappresentavano ragni (spider) e pipistrelli (bat), oltre a un barile di mele. L'obiettivo era eliminare tutti i ragni mantenendo intatti gli altri elementi. Ho usato il comando rm con un pattern con wildcard (\*spider\*) per cancellare in un colpo solo tutti i file il cui nome conteneva la parola "spider":

```

~
[mission 8] $ cd /
[mission 8] $ cd Castle
~/Castle
[mission 8] $ ls
Cellar Great_hall Main_building Main_tower Observatory

~/Castle
[mission 8] $ cd Cellar
~/Castle/Cellar
[mission 8] $ la
la: command not found

~/Castle/Cellar
[mission 8] $ ls
10516_spider_40 13701_spider_3 19092_spider_21 26167_spider_9 32678_spider_41 7298_spider_33
10526_spider_13 13960_spider_42 19931_spider_7 26188_spider_45 357_bat_1 7412_spider_5
11485_bat_2 14235_spider_22 20250_spider_28 26398_spider_11 5448_spider_46 919_spider_10
11638_spider_1 1429_spider_36 20945_spider_29 27902_spider_39 5602_spider_48 9514_spider_17
11753_spider_50 14826_spider_47 22162_spider_49 28791_spider_31 5860_spider_19 9941_spider_34
1179_spider_4 16224_spider_15 22838_spider_38 28996_spider_6 6539_spider_14 barrel_of_appl
12142_spider_43 16316_bat_5 24208_spider_18 29654_spider_25 6622_spider_23
12268_bat_3 16579_spider_44 24343_spider_30 29957_spider_27 6795_spider_24
12840_spider_8 16819_spider_20 24750_spider_26 30653_spider_2 6928_spider_37
13458_spider_12 17649_spider_16 25133_spider_32 31408_bat_4 7023_spider_35

~/Castle/Cellar
[mission 8] $ rm *spider*

~/Castle/Cellar
[mission 8] $ gsh check

Congratulations, mission 8 has been successfully completed!

[ progress was saved in /home/kali/gameshell-save.sh ]

| |
--+-----+--
| Use the command |
| $ gsh help |
| to get the list of "gsh" commands. |
--+-----+--
| |

```

## Missione 9 - I ragni nascosti

Proseguendo nella stessa location, ho scoperto che esistevano anche dei file "nascosti" (quelli che iniziano con un punto, visibili solo con `ls -a`) contenenti altri ragni. Ho quindi ripetuto l'operazione di pulizia con il pattern `.*spider*` per rimuovere anche questi file nascosti, completando la missione con successo. Il gioco ha inoltre sbloccato una piccola modifica al comportamento del comando `ls`, che da questo momento mostra un carattere `/` alla fine dei nomi delle directory:

```
~/Castle/Cellar
[mission 9] $ ls
11405_bat_2 12268_bat_3 16316_bat_5 31408_bat_4 357_bat_1 barrel_of_apples

~/Castle/Cellar
[mission 9] $ ls -la
.                .12875_spider_22 .18404_spider_40 .23948_spider_41 .30469_spider_45 .3629_spider_49
..               .13504_spider_35 .18938_bat_4    .24117_spider_6  .30694_spider_30 .4414_spider_37
.10674_spider_29 .13657_spider_18 .20375_spider_32 .24214_spider_17 31408_bat_4    .4475_spider_44
.11269_spider_38 .14064_bat_1     .20410_spider_50 .25090_spider_39 .31809_spider_33 .5568_bat_5
.11410_spider_31 .14543_spider_1  .20772_spider_10 .26156_spider_23 .32076_bat_3    .5704_spider_28
.11456_spider_25 .14657_spider_46 .21177_spider_26 .28134_spider_14 .32197_spider_3 .7814_spider_36
.11485_bat_2     .15469_spider_21 .21259_spider_19 .28310_spider_16 .32549_spider_27 .9930_spider_11
.11661_spider_12 .1568_spider_5   .22765_spider_43 .2845_spider_20  .32597_spider_48 barrel_of_apples
.11869_spider_13 .15703_spider_34 .23204_spider_8  .29131_spider_24 .32676_spider_4
.120_spider_47   16316_bat_5     .23263_spider_15 .29339_spider_7  .32739_spider_42
12268_bat_3     .17552_spider_9 .23638_spider_2  .30124_bat_2    357_bat_1

~/Castle/Cellar
[mission 9] $ rm .*spider*

~/Castle/Cellar
[mission 9] $ gsh check

Congratulations, mission 9 has been successfully completed!

      Congratulations !

      From now on, the ``ls`` command will automatically show a "/" character at the
      end of directories.
```

## Missione 12 - Recuperare il dipinto

Nell'ultima missione documentata mi sono spostato al primo piano della torre principale (Main\_tower/First\_floor), dove ho trovato tre dipinti (painting\_...) e la cartella Second\_floor. Ho individuato il dipinto richiesto e l'ho copiato, tramite il comando cp, nel baule della Capanna nella Foresta, completando così la missione:

```
[mission 12] $ ls
First_floor/

~/Castle/Main_tower
[mission 12] $ cd First_floor

~/Castle/Main_tower/First_floor
[mission 12] $ ls -l
total 16
-rw-rw-r-- 1 kali kali 1455 Apr  1 1998 painting_jtGPceSK
-rw-rw-r-- 1 kali kali 1054 Jan  4 2016 painting_YORCzYl
-rw-rw-r-- 1 kali kali 1503 Jan  1 1984 painting_YKcxBiT
drwxrwxr-x 3 kali kali 4096 Sep  4 13:53 Second_floor/

~/Castle/Main_tower/First_floor
[mission 12] $ cat ls
cat: ls: No such file or directory

~/Castle/Main_tower/First_floor
[mission 12] $ ls
painting_jtGPceSK  painting_YORCzYl  painting_YKcxBiT  Second_floor/

~/Castle/Main_tower/First_floor
[mission 12] $ cp painting_YKcxBiT ~/Forest/Hut/Chest/

~/Castle/Main_tower/First_floor
[mission 12] $ gsh check

Congratulations, mission 12 has been successfully completed!

[ progress was saved in /home/kali/gameshell-save.sh ]

+-----+
| Use the command |
|   $ gsh help   |
| to get the list of "gsh" commands. |
+-----+
```

## 6. Stato di avanzamento

Al momento della stesura di questo report ho documentato con successo, tramite screenshot, le missioni 1, 2, 6, 8, 9 e 12 di GameShell.

## Conclusioni

Attraverso questa pratica, per la parte svolta finora, ho imparato a:

- Installare e avviare correttamente GameShell su Kali Linux, gestendo anche un errore di rete durante il download
- Utilizzare i comandi di base della shell: cd, pwd, ls, mv, cp, rm
- Usare i pattern con wildcard (\*) per selezionare più file contemporaneamente
- Gestire i file nascosti (dot files) con ls -a e rimuoverli con pattern dedicati
- Usare gli strumenti specifici del gioco (gsh goal, gsh check, gsh help) per orientarsi tra le missioni
- Navigare una struttura di cartelle annidate per raggiungere obiettivi specifici e spostare/copiare file tra location diverse

Proseguirò con le missioni successive per consolidare ulteriormente la padronanza della riga di comando.